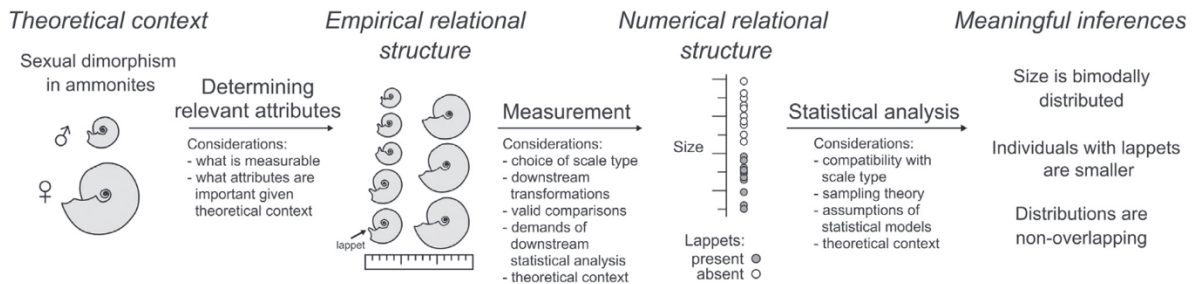


Assignment #5



Theoretical Context

The theoretical context to the research I am conducting is regarding perseverance and its impact following a failure. I am looking at it while considering the factor of GRIT (perseverance), motivation, affect changes and loser effects. Perseverance can be described as persistence in a course of action despite failure and no indication of future success. Previous research has established the existence of winner and loser effects amongst humans (Smith and Dukas 2024a). Winner and loser effects can be described as a phenomenon where winners are more likely to win subsequent matches while contest losers are more likely to lose in their next fights. Additional research identified affective changes as the proximate mechanism behind winner and loser effects (Smith and Dukas 2024b). Furthermore, previous studies showcased that undergoing a loss can result in an individual wanting to switch to a different competitive venue instead of trying again within the same platform (Smith and Dukas 2025). With this in mind, we are interested in understanding whether perseverance can influence an individual's decision to want to try again or switch when put in a losing condition. Additionally, we are interested in the proximate mechanism behind how perseverance might influence an individual to want to try again. Specifically, how perseverance could mitigate the affective changes that come with a loss.

Based on the measurement theory and theoretical background, the relevant and measurable attributes are the perseverance levels, affective changes, and motivation behind their choices. The perseverance levels can be measured using the GRIT scale, affective changes using the PANAS scale, motivation using the two interest questionnaires, and decision to switch or keep playing the same game using a question with binary options.

Scale and measurement

The GRIT-S (GRIT-Short) scale comprises of 8 Likert type items. The responses are ordinal since they are ranked in a particular order. However, when the items are summed up or averaged, the cumulative GRIT score is treated as an interval scale measurement for statistical analysis, with a higher numerical score representing a more perseverant individual. This makes the GRIT score a continuous variable. The PANAS scale also known as the Positive and Negative Affect Schedule consists of 20 Likert type questions. The items are ordinal, but the summed-up score is treated similarly to that of

GRIT where it is treated as an interval scale. Both the GRIT and PANAS scales will be used as a continuous variable during analysis. This allows for interpretations with higher and lower affective changes. Moreover, the 2 videogame motivation questions are ordinal scales with one of them being a Likert scale. Both questions will be used in a categorical way during statistical analysis. Finally, the individual's preference to switch will be in a binary format which is as a nominal scale. The participants get to either keep playing the original game they played in round 1 or they get to switch to a different racing game in round 2. This question will be analyzed as a categorical variable.

P.S I wanted to share my survey so you guys would be able to see it and get a better idea, but I am not too sure how to share a view only version of my survey

Statistical Analysis

The primary outcome variable we are looking at is the binary nominal variable where the participants get to choose what they would like to do. We believe the primary predictor for what someone would do is their GRIT score. Since the summer up GRIT score is a continuous variable, I believe I can conduct a logistic regression analysis, which models the probability of persevering and wanting to try again as a function of GRIT, mood and interest in the videogame. Additionally, I will also be looking into running a comparison test to investigate whether there is a difference in GRIT scores amongst individuals who chose to switch to a different videogame versus individuals who decided to keep playing the same game. I believe an independent t-test would suffice in this situation. Moreover, I believe I can conduct a Chi square test of independence to investigate whether an individual's decision to switch or stay could be compared to their motivation behind playing videogames. I am using the Chi square test here since both the variables I am interested in are categorical variables. Finally, I plan on conducting a correlational Pearson's r test between GRIT scores and affective changes, since these 2 variables are continuous. This correlational test will tell me if there is a correlational association between these two variables. I am predicting that higher GRIT scores will result in less negative affective decline, allowing individuals to persevere and try again after a loss.

What counts as important difference?

The one comparison I am choosing to investigate is the independent t-test. This comparison will tell me if there is a statistical difference in the GRIT scores amongst participants who choose to switch to a different game versus participants who chose to stay playing the same game. I plan on utilizing the material learned in class to understand if the difference found is important. As mentioned in class, a statistical significance due to a low p value does not provide us with the needed information to understand whether the difference is important. A statistical significance gives us clarity on seeing that there is indeed a difference. To make a meaningful distinction on whether the difference that was found is important or not, I will need to look for confidence intervals and effect sizes. If the confidence intervals spans both positive and negative values, the estimated difference in GRIT scores will be interpreted as inconclusive with respect to the direction of the effect between the two groups. In this case, a statistically

significant p-value indicates evidence against the null hypothesis but does not specify the magnitude or practice importance of the effect.

References:

Smith N, Dukas R. 2024a. Winner and loser effects in humans: evidence from randomized trials. *Animal Behaviour*. 207:101–107.
<https://doi.org/10.1016/j.anbehav.2023.10.017>

Smith N, Dukas R. 2024b. Winner and Loser Effects and Social Rank In Humans. *The Quarterly Review of Biology*. 99(3):157–174. <https://doi.org/10.1086/732049>

Smith N, Dukas R. 2025. Adaptive significance of winner and loser effects: rank-dependent optimal behaviour. *Animal Behaviour*. 223:123184.
<https://doi.org/10.1016/j.anbehav.2025.123184>